PG No. 45 $C_{6h}(c)$ $6/m$ [hexagonal] (internal axial dipole)

* Harmonics for rank 0

$\vec{\mathbb{G}}_0^{(1,1)}[g](A_u)$

** symmetry

1

** expression

$$\frac{\sqrt{3}G_x x}{3}+\frac{\sqrt{3}G_y y}{3}+\frac{\sqrt{3}G_z z}{3}$$

* Harmonics for rank 1

$\vec{\mathbb{G}}_1^{(1,-1)}[g](A_g)$

** symmetry

z

** expression

G_z

$\vec{\mathbb{G}}_1^{(1,1)}[g](A_g)$

** symmetry

z

** expression

$$\frac{3\sqrt{10}G_x xz}{10}+\frac{3\sqrt{10}G_y yz}{10}-\frac{\sqrt{10}G_z\left(x^2+y^2-2z^2\right)}{10}$$

$\vec{\mathbb{G}}_1^{(1,-1)}[g](E_{1g}^{(a)})$

** symmetry

$$\frac{\sqrt{2}x}{2}+\frac{\sqrt{2}iy}{2}$$

** expression

$$\frac{\sqrt{2}G_x}{2}+\frac{\sqrt{2}iG_y}{2}$$

$\vec{\mathbb{G}}_1^{(1,1)}[g](E_{1g}^{(a)})$

** symmetry

$$\frac{\sqrt{2}x}{2}+\frac{\sqrt{2}iy}{2}$$

** expression

$$\frac{\sqrt{5}G_x\left(2x^2+3ixy-y^2-z^2\right)}{10}-\frac{\sqrt{5}iG_y\left(x^2+3ixy-2y^2+z^2\right)}{10}+\frac{3\sqrt{5}G_z z\left(x+iy\right)}{10}$$

$\vec{\mathbb{G}}_1^{(1,-1)}[g](E_{1g}^{(b)})$

** symmetry

$$\frac{\sqrt{2}x}{2}-\frac{\sqrt{2}iy}{2}$$

** expression

$$\frac{\sqrt{2}G_x}{2}-\frac{\sqrt{2}iG_y}{2}$$

$\vec{\mathbb{G}}_1^{(1,1)}[g](E_{1g}^{(b)})$

** symmetry

$$\frac{\sqrt{2}x}{2}-\frac{\sqrt{2}iy}{2}$$

** expression

$$\frac{\sqrt{5}G_x\left(2x^2-3ixy-y^2-z^2\right)}{10}+\frac{\sqrt{5}iG_y\left(x^2-3ixy-2y^2+z^2\right)}{10}+\frac{3\sqrt{5}G_z z\left(x-iy\right)}{10}$$

$\vec{\mathbb{Q}}_1^{(1,0)}[g](A_u)$

** symmetry

z

** expression

$$\frac{\sqrt{2}G_x y}{2} - \frac{\sqrt{2}G_y x}{2}$$

$\vec{\mathbb{Q}}_1^{(1,0)}[g](E_{1u}^{(a)})$
** symmetry

$$\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$$

** expression

$$-\frac{iG_x z}{2} + \frac{G_y z}{2} + \frac{iG_z\left(x+iy\right)}{2}$$

$\vec{\mathbb{Q}}_1^{(1,0)}[g](E_{1u}^{(b)})$
** symmetry

$$\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$$

** expression

$$\frac{iG_x z}{2} + \frac{G_y z}{2} - \frac{iG_z\left(x-iy\right)}{2}$$

* Harmonics for rank 2
$\vec{\mathbb{Q}}_2^{(1,0)}[g](A_g)$
** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

** expression

$$\frac{\sqrt{6}G_x yz}{2} - \frac{\sqrt{6}G_y xz}{2}$$

$\vec{\mathbb{Q}}_2^{(1,0)}[g](E_{1g}^{(a)})$
** symmetry

$$\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$$

** expression

$$\frac{G_x\left(xy+iy^2-iz^2\right)}{2} - \frac{G_y\left(x^2+ixy-z^2\right)}{2} + \frac{iG_z z\left(x+iy\right)}{2}$$

$\vec{\mathbb{Q}}_2^{(1,0)}[g](E_{1g}^{(b)})$
** symmetry

$$\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$$

** expression

$$\frac{G_x\left(xy-iy^2+iz^2\right)}{2} - \frac{G_y\left(x^2-ixy-z^2\right)}{2} - \frac{iG_z z\left(x-iy\right)}{2}$$

$\vec{\mathbb{Q}}_2^{(1,0)}[g](E_{2g}^{(a)})$
** symmetry

$$\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$$

** expression

$$\frac{iG_x z\left(x-iy\right)}{2} + \frac{G_y z\left(x-iy\right)}{2} - \frac{iG_z\left(x-iy\right)^2}{2}$$

$\vec{\mathbb{Q}}_2^{(1,0)}[g](E_{2g}^{(b)})$
** symmetry

$$\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$$

** expression

$$-\frac{iG_x z\left(x+iy\right)}{2} + \frac{G_y z\left(x+iy\right)}{2} + \frac{iG_z\left(x+iy\right)^2}{2}$$

$\vec{\mathbb{G}}_2^{(1,-1)}[g](A_u)$
** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{\sqrt{6}G_x x}{6}-\frac{\sqrt{6}G_y y}{6}+\frac{\sqrt{6}G_z z}{3}$$

$\vec{\mathbb{G}}_2^{(1,1)}[g](A_u)$
** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{\sqrt{21}G_x x\left(x^2+y^2-4z^2\right)}{14}-\frac{\sqrt{21}G_y y\left(x^2+y^2-4z^2\right)}{14}-\frac{\sqrt{21}G_z z\left(3x^2+3y^2-2z^2\right)}{14}$$

$\vec{\mathbb{G}}_2^{(1,-1)}[g](E_{1u}^{(a)})$
** symmetry

$$\frac{\sqrt{6}xz}{2}+\frac{\sqrt{6}iyz}{2}$$

** expression

$$\frac{G_x z}{2}+\frac{iG_y z}{2}+\frac{G_z\left(x+iy\right)}{2}$$

$\vec{\mathbb{G}}_2^{(1,1)}[g](E_{1u}^{(a)})$
** symmetry

$$\frac{\sqrt{6}xz}{2}+\frac{\sqrt{6}iyz}{2}$$

** expression

$$\frac{\sqrt{14}G_x z\left(4x^2+5ixy-y^2-z^2\right)}{14}-\frac{\sqrt{14}iG_y z\left(x^2+5ixy-4y^2+z^2\right)}{14}-\frac{\sqrt{14}G_z\left(x+iy\right)\left(x^2+y^2-4z^2\right)}{14}$$

$\vec{\mathbb{G}}_2^{(1,-1)}[g](E_{1u}^{(b)})$
** symmetry

$$\frac{\sqrt{6}xz}{2}-\frac{\sqrt{6}iyz}{2}$$

** expression

$$\frac{G_x z}{2}-\frac{iG_y z}{2}+\frac{G_z\left(x-iy\right)}{2}$$

$\vec{\mathbb{G}}_2^{(1,1)}[g](E_{1u}^{(b)})$
** symmetry

$$\frac{\sqrt{6}xz}{2}-\frac{\sqrt{6}iyz}{2}$$

** expression

$$\frac{\sqrt{14}G_x z\left(4x^2-5ixy-y^2-z^2\right)}{14}+\frac{\sqrt{14}iG_y z\left(x^2-5ixy-4y^2+z^2\right)}{14}-\frac{\sqrt{14}G_z\left(x-iy\right)\left(x^2+y^2-4z^2\right)}{14}$$

$\vec{\mathbb{G}}_2^{(1,-1)}[g](E_{2u}^{(a)})$
** symmetry

$$\frac{\sqrt{6}x^2}{4}-\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$$

** expression

$$\frac{G_x\left(x-iy\right)}{2}-\frac{iG_y\left(x-iy\right)}{2}$$

$\vec{\mathbb{G}}_2^{(1,1)}[g](E_{2u}^{(a)})$
** symmetry

$$\frac{\sqrt{6}x^2}{4}-\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$$

** expression

$$\frac{\sqrt{14}G_x\left(x-iy\right)\left(3x^2-5ixy-2y^2-2z^2\right)}{28}+\frac{\sqrt{14}iG_y\left(x-iy\right)\left(2x^2-5ixy-3y^2+2z^2\right)}{28}+\frac{5\sqrt{14}G_z z\left(x-iy\right)^2}{28}$$

$\vec{\mathbb{G}}_2^{(1,-1)}[g](E_{2u}^{(b)})$

** symmetry

$$\frac{\sqrt{6}x^2}{4}+\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$$

** expression

$$\frac{G_x\left(x+iy\right)}{2}+\frac{iG_y\left(x+iy\right)}{2}$$

$\vec{\mathbb{G}}_2^{(1,1)}[g](E_{2u}^{(b)})$

** symmetry

$$\frac{\sqrt{6}x^2}{4}+\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$$

** expression

$$\frac{\sqrt{14}G_x\left(x+iy\right)\left(3x^2+5ixy-2y^2-2z^2\right)}{28}-\frac{\sqrt{14}iG_y\left(x+iy\right)\left(2x^2+5ixy-3y^2+2z^2\right)}{28}+\frac{5\sqrt{14}G_z z\left(x+iy\right)^2}{28}$$

* Harmonics for rank 3

$\vec{\mathbb{G}}_3^{(1,-1)}[g](A_g)$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{15}G_xxz}{5}-\frac{\sqrt{15}G_yyz}{5}-\frac{\sqrt{15}G_z\left(x^2+y^2-2z^2\right)}{10}$$

$\vec{\mathbb{G}}_3^{(1,1)}[g](A_g)$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{5G_xxz\left(3x^2+3y^2-4z^2\right)}{12}-\frac{5G_yyz\left(3x^2+3y^2-4z^2\right)}{12}+\frac{G_z\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{12}$$

$\vec{\mathbb{G}}_3^{(1,-1)}[g](B_g,1)$

** symmetry

$$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$$

** expression

$$\frac{\sqrt{6}G_xxy}{2}+\frac{\sqrt{6}G_y\left(x-y\right)\left(x+y\right)}{4}$$

$\vec{\mathbb{G}}_3^{(1,-1)}[g](B_g,2)$

** symmetry

$$\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$$

** expression

$$\frac{\sqrt{6}G_x\left(x-y\right)\left(x+y\right)}{4}-\frac{\sqrt{6}G_yxy}{2}$$

$\vec{\mathbb{G}}_3^{(1,1)}[g](B_g,1)$

** symmetry

$$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$$

** expression

$$\frac{\sqrt{10}G_xxy\left(15x^2-13y^2-6z^2\right)}{24}-\frac{\sqrt{10}G_y\left(3x^4-21x^2y^2+3x^2z^2+4y^4-3y^2z^2\right)}{24}+\frac{7\sqrt{10}G_yyz\left(3x^2-y^2\right)}{24}$$

$\vec{\mathbb{G}}_3^{(1,1)}[g](B_g, 2)$

** symmetry

$$\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$$

** expression

$$\frac{\sqrt{10}G_x\left(4x^4 - 21x^2y^2 - 3x^2z^2 + 3y^4 + 3y^2z^2\right)}{24} + \frac{\sqrt{10}G_y xy\left(13x^2 - 15y^2 + 6z^2\right)}{24} + \frac{7\sqrt{10}G_z xz\left(x^2 - 3y^2\right)}{24}$$

$\vec{\mathbb{G}}_3^{(1,-1)}[g](E_{1g}^{(a)})$

** symmetry

$$-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$$

** expression

$$-\frac{\sqrt{5}G_x\left(3x^2 + 2ixy + y^2 - 4z^2\right)}{20} - \frac{\sqrt{5}iG_y\left(x^2 - 2ixy + 3y^2 - 4z^2\right)}{20} + \frac{2\sqrt{5}G_z z\left(x + iy\right)}{5}$$

$\vec{\mathbb{G}}_3^{(1,1)}[g](E_{1g}^{(a)})$

** symmetry

$$-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$$

** expression

$$-\frac{\sqrt{3}G_x\left(4x^4 + 5ix^3y + 3x^2y^2 - 27x^2z^2 + 5ixy^3 - 30ixyz^2 - y^4 + 3y^2z^2 + 4z^4\right)}{24} + \frac{\sqrt{3}iG_y\left(x^4 + 5ix^3y - 3x^2y^2 - 3x^2z^2 + 5ixy^3 - 30ixyz^2 - 4y^4 + 27y^2z^2 - 4z^4\right)}{24} - \frac{5\sqrt{3}G_z z\left(x + iy\right)\left(3x^2 + 3y^2 - 4z^2\right)}{24}$$

$\vec{\mathbb{G}}_3^{(1,-1)}[g](E_{1g}^{(b)})$

** symmetry

$$-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$$

** expression

$$-\frac{\sqrt{5}G_x\left(3x^2 - 2ixy + y^2 - 4z^2\right)}{20} + \frac{\sqrt{5}iG_y\left(x^2 + 2ixy + 3y^2 - 4z^2\right)}{20} + \frac{2\sqrt{5}G_z z\left(x - iy\right)}{5}$$

$\vec{\mathbb{G}}_3^{(1,1)}[g](E_{1g}^{(b)})$

** symmetry

$$-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$$

** expression

$$-\frac{\sqrt{3}G_x\left(4x^4 - 5ix^3y + 3x^2y^2 - 27x^2z^2 - 5ixy^3 + 30ixyz^2 - y^4 + 3y^2z^2 + 4z^4\right)}{24} - \frac{\sqrt{3}iG_y\left(x^4 - 5ix^3y - 3x^2y^2 - 3x^2z^2 - 5ixy^3 + 30ixyz^2 - 4y^4 + 27y^2z^2 - 4z^4\right)}{24} - \frac{5\sqrt{3}G_z z\left(x - iy\right)\left(3x^2 + 3y^2 - 4z^2\right)}{24}$$

$\vec{\mathbb{G}}_3^{(1,-1)}[g](E_{2g}^{(a)})$

** symmetry

$$\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{\sqrt{2}G_x z\left(x - iy\right)}{2} - \frac{\sqrt{2}iG_y z\left(x - iy\right)}{2} + \frac{\sqrt{2}G_z\left(x - iy\right)^2}{4}$$

$\vec{\mathbb{G}}_3^{(1,1)}[g](E_{2g}^{(a)})$

** symmetry

$$\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{\sqrt{30}G_x z\left(x - iy\right)\left(5x^2 - 7ixy - 2y^2 - 2z^2\right)}{24} + \frac{\sqrt{30}iG_y z\left(x - iy\right)\left(2x^2 - 7ixy - 5y^2 + 2z^2\right)}{24} - \frac{\sqrt{30}G_z\left(x - iy\right)^2\left(x^2 + y^2 - 6z^2\right)}{24}$$

$\vec{\mathbb{G}}_3^{(1,-1)}[g](E_{2g}^{(b)})$
** symmetry

$$\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{\sqrt{2}G_xz\left(x+iy\right)}{2}+\frac{\sqrt{2}iG_yz\left(x+iy\right)}{2}+\frac{\sqrt{2}G_z\left(x+iy\right)^2}{4}$$

$\vec{\mathbb{G}}_3^{(1,1)}[g](E_{2g}^{(b)})$
** symmetry

$$\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{\sqrt{30}G_xz\left(x+iy\right)\left(5x^2+7ixy-2y^2-2z^2\right)}{24}-\frac{\sqrt{30}iG_yz\left(x+iy\right)\left(2x^2+7ixy-5y^2+2z^2\right)}{24}-\frac{\sqrt{30}G_z\left(x+iy\right)^2\left(x^2+y^2-6z^2\right)}{24}$$

$\vec{\mathbb{Q}}_3^{(1,0)}[g](A_u)$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{3}G_xy\left(x^2+y^2-4z^2\right)}{4}+\frac{\sqrt{3}G_yx\left(x^2+y^2-4z^2\right)}{4}$$

$\vec{\mathbb{Q}}_3^{(1,0)}[g](B_u,1)$
** symmetry

$$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$$

** expression

$$-\frac{\sqrt{30}G_xz\left(x-y\right)\left(x+y\right)}{8}+\frac{\sqrt{30}G_yxyz}{4}+\frac{\sqrt{30}G_zx\left(x^2-3y^2\right)}{8}$$

$\vec{\mathbb{Q}}_3^{(1,0)}[g](B_u,2)$
** symmetry

$$\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$$

** expression

$$\frac{\sqrt{30}G_xxyz}{4}+\frac{\sqrt{30}G_yz\left(x-y\right)\left(x+y\right)}{8}-\frac{\sqrt{30}G_zy\left(3x^2-y^2\right)}{8}$$

$\vec{\mathbb{Q}}_3^{(1,0)}[g](E_{1u}^{(a)})$
** symmetry

$$-\frac{\sqrt{3}x^3}{4}-\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2-\frac{\sqrt{3}iy^3}{4}+\sqrt{3}iyz^2$$

** expression

$$\frac{iG_xz\left(x^2-10ixy+11y^2-4z^2\right)}{8}-\frac{G_yz\left(11x^2+10ixy+y^2-4z^2\right)}{8}-\frac{iG_z\left(x+iy\right)\left(x^2+y^2-4z^2\right)}{8}$$

$\vec{\mathbb{Q}}_3^{(1,0)}[g](E_{1u}^{(b)})$
** symmetry

$$-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$$

** expression

$$-\frac{iG_xz\left(x^2+10ixy+11y^2-4z^2\right)}{8}-\frac{G_yz\left(11x^2-10ixy+y^2-4z^2\right)}{8}+\frac{iG_z\left(x-iy\right)\left(x^2+y^2-4z^2\right)}{8}$$

$\vec{\mathbb{Q}}_3^{(1,0)}[g](E_{2u}^{(a)})$
** symmetry

$$\frac{\sqrt{30}x^2z}{4}-\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{\sqrt{10}G_x\left(x-iy\right)\left(xy-iy^2+2iz^2\right)}{8}-\frac{\sqrt{10}G_y\left(x-iy\right)\left(x^2-ixy-2z^2\right)}{8}-\frac{\sqrt{10}iG_zz\left(x-iy\right)^2}{4}$$

$\vec{\mathbb{Q}}_3^{(1,0)}[g](E_{2u}^{(b)})$

** symmetry

$$\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{\sqrt{10}G_x\left(x+iy\right)\left(xy+iy^2-2iz^2\right)}{8}-\frac{\sqrt{10}G_y\left(x+iy\right)\left(x^2+ixy-2z^2\right)}{8}+\frac{\sqrt{10}iG_zz\left(x+iy\right)^2}{4}$$